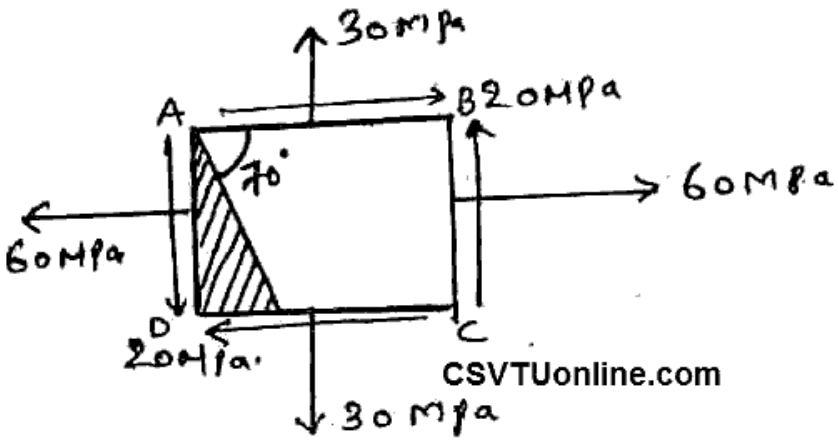

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Or

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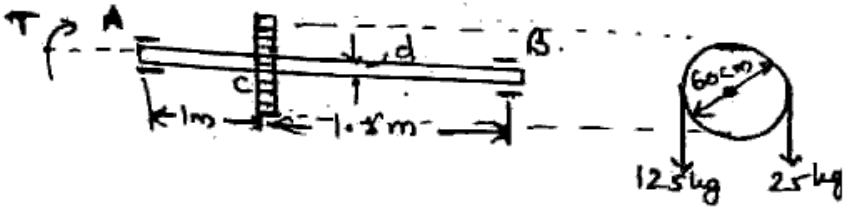
(c) A steel shaft supported in bearings A and B at its ends carries a pulley 60 cm diameter at C as shown in figure below. Power is applied by a torque T at end A. Tension in two branches of the belt being 125 kg and 25 kg. Find the required diameter 'd' for the shaft if working stresses are $\sigma_w = 850$ kg/cm^2 and $\tau_w = 425$ kg/cm^2. 14



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B. E. (Third Semester) Examination, April-May 2016

(New Scheme)

(Mech. Engg. Branch)

MECHANICS of SOLIDS-I

Time Allowed : Three hours

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Maximum Marks : 80

Minimum Pass Marks : 28

Note : Attempt all the questions. Part (a) is compulsory from each question. Attempt any two part from (b), (c) and (d) from Q. no. 1, 3 & 4 and any one part from Q. no. 2 & 5. Assume any suitable data if missing. The symbols used have usual meaning as in standard texts.

1. (a) If Young's modulus of elasticity of steel is 200 GPa and Poisson's ratio is 0·25. Find modulus of rigidity. 2

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(b) In the arrangement shown below, the 4 mm diameter cable BC is made of steel with E = 200 GPa.

Knowing that maximum stress in the cable must not exceed 190 MPa and that the elongation of the cable must not exceed 6 mm. Find the maximum load P that can be applied. 7

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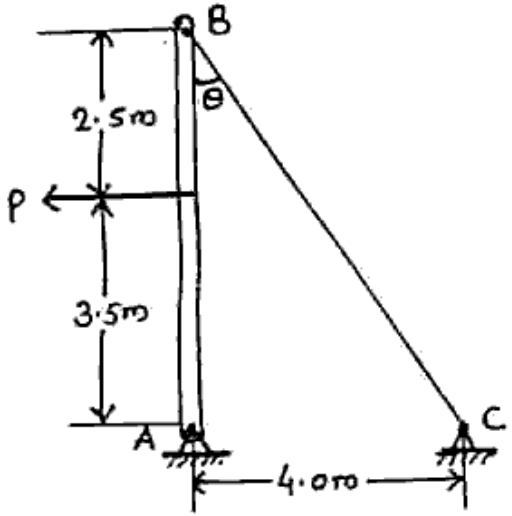

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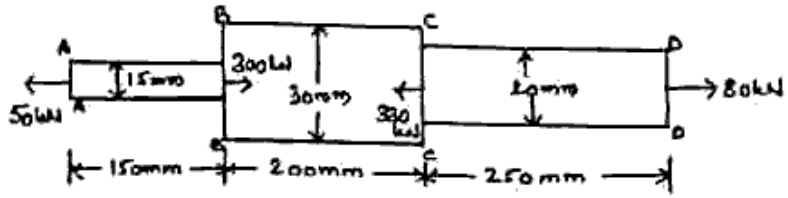
(c) A steel circular bar has three segments as shown in figure below. Determine :

(i) Total elongation of bar or net elongation.

(ii) Length of middle segment to have zero elongation of bar.

(should be solved as case-II).

(iii) The diameter of the last segment to have zero elongation of the bar. (should be solved as case III). Take $E = 205$ GPa. 7



(d) The aluminium shell if fully bonded to the brass core as shown in figure below. The assembly is unstressed at a temperature of 26°C. Considering only axial deformations, determine the stress when the temperature reaches 82°C (i) in the brass core (ii) in the aluminium shell. Take : 7

$$E_b = 103 \text{ GPa}, \ \alpha_b = 20 \cdot 9 \times 10^{-6} / °C$$

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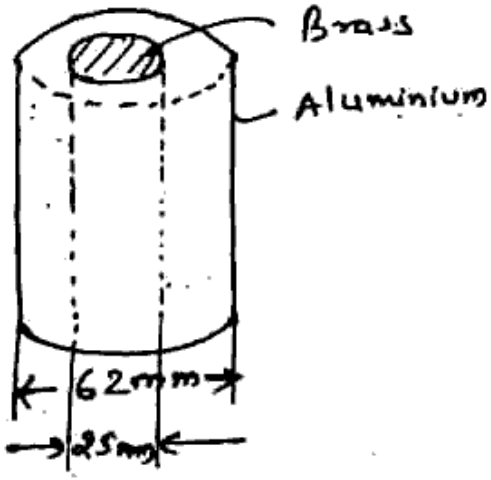
$$E_{al} = 73 \text{ GPa}, \ \alpha_{al} = 23 \cdot 2 \times 10^{-6} / °C$$



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2. (a) Fill in the blanks : 2

(i) Shear stress variation in beams of rectangular section is ..... in nature.

(ii) $(\tau)_{max}$ for circular cross section = ...

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times $(\tau_{average})$ for circular.

(b) A simply supported beam has a span of 8 m and carries a UDL of 20 kN/m over the whole span along with two point loads of 30 kN and 40 kN at 6 W and 7·0 m respectively from left hand support.

(i) Draw the beam with loading mentioned.

(ii) Draw the shear force and bending moment diagrams showing all the principal values. 14

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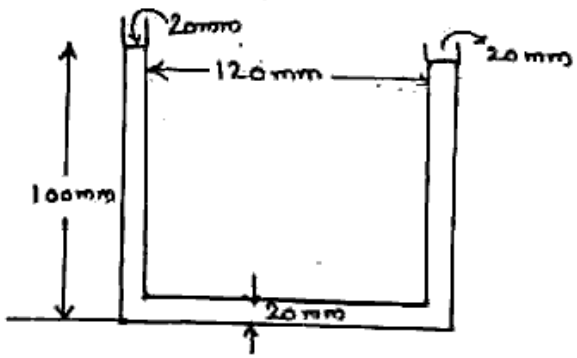
(c) The horizontal beam of section shown in fig. below is 4 m long and is simply supported at the ends. Calculate the maximum UDL it can carry if tensile and compressive stress must not exceed 25 MPa and 45 MPa respectively. 14



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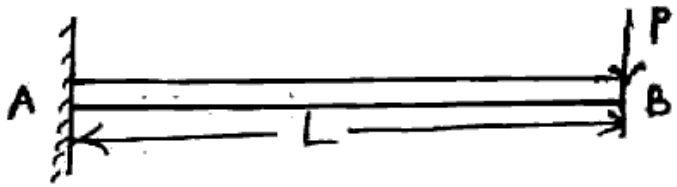
3. (a) Write the expression of slope and deflection of a cantilever beam of span 'L' and carrying a UDL of 'W' kN/m^2 at free end. 2

(b) Determine the slope and deflection of a cantilever beam at free end shown below by method of direct integration. 7



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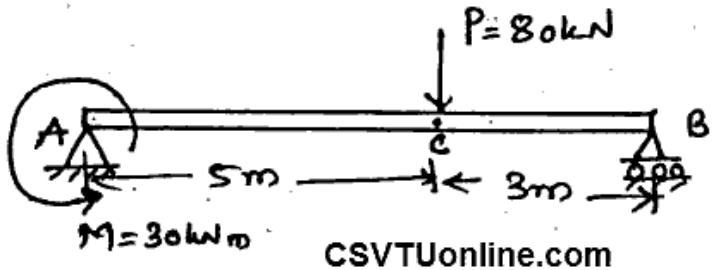
(c) Using Macaulay's method, determine the slope at end A and deflection at point C for beam loaded and supported as shown below : 7

Take E = 210 GPa and $I = 305 \times 10^6$ mm^4.



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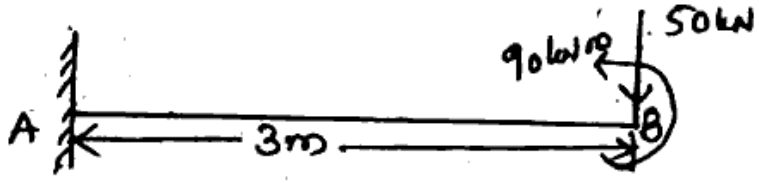
(d) Using Moment area method, determine the slope and deflection of end B of cantilever beam AB, when it is loaded as shown in figure below. Take $EI = 10 \times 10^6$ N-m^2. 7



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4. (a) Define Polar Moment of Inertia. 2

(b) A solid circular shaft transmit 75 kW at 200 rpm. Calculate the shaft diameter if the twist in the shaft is not to exceed 1° in 2 m length of the shaft and shear stress is limited to 50 MPa.
Take G = 100 GPa. 7

(c) Two shafts of the same material and same length are subjected to the same torque. If the first shaft is solid circular section, and second shaft is of hollow circular section, whose internal diameter is 2/3rd the

outside diameter and maximum shear stress developed in each shaft is same. Compare the weights of two shafts? 7

(d) A railway wagon weighing 40 kN and moving with a speed of 8 km/hr, is stopped by a buffer of 4 springs whose allowable maximum compression is 150 mm. Find out the number of turns in each spring, if the diameter of spring wire is 14 mm and diameter of coil is 80 mm. Take G = 84 GPa. 7

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5. (a) What are Principal Stresses? 2

(b) For the state of plane stress as shown below, determine analytically or graphically : 7

(i) Normal and shearing stress exerted on oblique face of shaded triangle.

(ii) Location of Principal planes.

(iii) Magnitude of Principal stresses.

(iv) Maximum shearing stress and corresponding normal stress in the plane of maximum shear. 14

C: